

Pre-Processing Fixes (Data Fixes)

This step focuses on fixing bias that already exists in the data before the model is trained. It ensures the dataset fairly represents all users and that the model is not learning from flawed or misleading inputs. This stage applies only when causal analysis identifies bias at the data or measurement stage.

This stage must only be applied when causal fairness analysis identifies bias originating at the data or measurement stage.

When to Use

Apply this step when fairness issues are traced back to the data itself, such as:

- Certain groups are under-represented, causing the model to perform poorly for them
- Lower data quality for specific users or regions
- Biased labels or measurements that reflect historical or human bias rather than reality

When Not to Use

Pre-processing should not be applied when:

- Bias originates from model behavior rather than data
- Data is already balanced and representative
- Measurement processes are consistent across groups

Applying pre-processing in these scenarios may distort valid signals and introduce new bias.

Allowed Actions

Teams may:

- Rebalance the dataset (e.g., sampling or weighting) with a clear justification
- Remove low-quality or corrupted records that skew outcomes
- Normalize features consistently across all groups
- Correct known data errors based on verified sources

Required Output

Every change must be documented through:

- A Data Bias Summary explaining what was wrong and why fixes were needed
- A Dataset Change Log detailing what was changed, when, and by whom

Clear documentation prevents ad-hoc decisions and enables future audits and reviews. Completion of this stage requires validation before proceeding to governance approval.

Connection to Intervention Selection and Validation

This stage is selected based on the Risk-Based Intervention Framework when bias is identified at the data stage.

All pre-processing interventions must be validated to ensure:

- Fairness improvement is measurable
- Model performance remains acceptable
- No new bias is introduced

No data intervention is considered complete without validation approval.